

LOSS-INFORMED TRANSFORMER TRAINING WITH LEARNED RATIONAL ACTIVATIONS

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ABSTRACT

We introduce a joint activation–optimizer design that achieves lower mean validation loss than AdamW, Muon, and SOAP in Transformer pretraining at matched training-token budgets. Our approach uses the model’s learned nonlinear responses to guide both the construction of matrix updates and their joint allocation across groups and layers. **GRAIN**, Groupwise Rational Activation with Internal Normalization, learns a rational activation function for each feature group in normalized coordinates, allowing different groups to adapt their nonlinear response shapes during training. **TILLER**, Tangent-Informed Loss Ledger for Equal-budget Reweighting, uses these learned responses to shape matrix updates and estimate their first-order effects on the loss. Measuring these effects jointly reveals how the loss-response patterns of different updates reinforce or offset one another, information that their individual average descent scores do not capture. TILLER maintains a running summary of these cross-update relationships and combines it with current gradients to determine the relative strengths of updates across groups and layers. The gains hold across every tested corpus and model size with both SwiGLU and GRAIN baseline models, demonstrating improvements beyond activation replacement alone.

1 INTRODUCTION

Learning an activation function gives a Transformer control over both its feature representation and the response of those features to weight updates. Fixed nonlinearities such as GELU and the SiLU gate in SwiGLU provide effective transformations (Hendrycks & Gimpel, 2016; Shazeer, 2020). Rational activations make the response shape trainable (Molina et al., 2020), with approximation advantages that motivate their use as expressive neural representations (Boullé et al., 2020; Tang & Townsend, 2026). Group-shared rational functions bring this flexibility to Transformer feed-forward blocks (Yang & Wang, 2025). The changing response also supplies information for optimizing the matrices around it: learning the nonlinearity changes how an update propagates through the network.

The incoming and outgoing projections expose two complementary paths. An incoming update changes features through the activation Jacobian; an outgoing update recombines the activated features. Relating these changes to the downstream loss gives a functional description of each matrix direction. The interaction between directions is equally important. Two groups can have the same average descent signal while their loss responses reinforce or offset one another at shared activation positions. Separate descent scores leave these relationships unresolved. Measuring the responses together makes them available for choosing the strengths of simultaneous updates.

We introduce GRAIN, Groupwise Rational Activation with Internal Normalization, as the learned representation for this construction. Each feature group evaluates its own rational function on RMS-normalized coordinates and multiplies the result by the same normalization scale (Zhang & Sennrich, 2019). The learned coefficients control the response to relative feature magnitudes, while the output retains a dependence on input amplitude. Its Jacobian describes both the local rational slope and the effect of changing the group scale. The features and their derivatives provide the two response paths used to guide optimization.

Matrix-aware optimization uses matrix structure and activation sensitivity to shape updates (Martens & Grosse, 2015; Vyas et al., 2025; Liu et al., 2025). TILLER, Tangent-Informed Loss Ledger for Equal-budget Reweighting, uses the learned responses to construct matrix directions and coordinate

their joint effect on the loss. It uses the response geometry to rotate an orthogonalized momentum direction along a factorized adaptive tangent, with orientation selected by the gradient (Shazeer & Stern, 2018). It then records the selected feed-forward directions’ loss responses at shared positions in a compact temporal ledger. Together with the current gradients, this history defines a joint model for allocating update strength across groups and layers. The allocation preserves the combined norm of that step’s feed-forward directions, so the ledger changes where the update acts according to the responses it produces.

This construction gives two training methods. Full TILLER continually adapts directions and their joint allocation throughout training. The staged method applies TILLER for an initial interval, then continues from the learned model with a chosen optimizer. GRAIN remains trainable in both methods. Full TILLER maintains loss-response coordination along the complete trajectory; staged training combines a coordinated training stage with the update rule and computational cost of its continuation optimizer.

Across five pretraining corpora and both evaluated model sizes, GRAIN with full TILLER achieves the lowest mean validation loss among the tested activation–optimizer configurations at matched token budgets. The comparisons include AdamW, Muon, SOAP, and factorized adaptive optimizers. At the larger model size, the evaluated staged configuration also achieves lower mean validation loss than every baseline, while using less total training time than full TILLER on every corpus.

[TODO: Experiment section]

2 RELATED WORK

Padé Activation Units learn rational curves jointly with network weights (Molina et al., 2020). Kolmogorov–Arnold Transformer (KAT) implements group-shared rational functions followed by linear maps; its reported configuration shares numerator coefficients across groups (Yang & Wang, 2025). GRAIN evaluates each group’s rational function in RMS-normalized coordinates (Zhang & Sennrich, 2019) and restores the group scale at its output. It retains group sharing while learning both numerator and denominator coefficients independently for each group.

Approximation theory establishes parameter-efficiency separations between rational and ReLU networks (Boullé et al., 2020). *Rational Neural Networks have Expressivity Advantages*, by Maosen Tang and Alex Townsend, extends these comparisons to bounded-parameter smooth-activation networks and analyzes coefficient redundancy under learned affine normalization (Tang & Townsend, 2026). GRAIN uses an internal normalization–rescaling map with no additional learned affine transform.

Muon orthogonalizes momentum (Jordan et al., 2024), while Adafactor estimates gradient second moments from row and column statistics (Shazeer & Stern, 2018). TILLER combines these direction constructions using the learned activation response. Loss-weighted features and Jacobians determine the extent of a rotation from the polar direction along an adaptive tangent, with its orientation selected by the current gradient.

Joint update models differ in their search directions and the information used to compare them. K-FAC constructs parameter-space Fisher approximations from activation and backpropagated-derivative moments, including cross-layer blocks (Martens & Grosse, 2015). Newton–Muon derives input-second-moment preconditioning from a matrix quadratic surrogate (Du & Su, 2026). Krylov Subspace Descent optimizes the nonlinear objective over a curvature-generated Krylov subspace (Vinyals & Povey, 2012). TILLER places its quadratic model on coefficients multiplying selected pairs of incoming and outgoing group directions. Current gradients supply the linear term. A discounted history of shared-position response probes, augmented between probes by current-gradient information, supplies the quadratic penalty. Its cross terms couple coefficients across groups and layers; the allocation preserves the combined feed-forward direction norm before weight decay.

Optimizer switching combines update rules along a training trajectory. SWATS selects an Adam-to-SGD transition using a projection-based trigger (Keskar & Socher, 2017). Our staged method applies TILLER until a chosen step, then continues with a selected optimizer while retaining the learned parameters and trainable GRAIN activations.

3 METHOD

We introduce GRAIN in Transformer feed-forward blocks and TILLER to coordinate matrix updates using GRAIN’s loss response. We first define the block, then the update shared by full-horizon and staged training.

3.1 GRAIN

GRAIN (Groupwise Rational Activation with Internal Normalization) replaces a fixed nonlinearity with group-shared rational functions. For input $x \in \mathbb{R}^d$, partition the intermediate features into G groups of width w . The incoming and outgoing projections of group g are $A_g \in \mathbb{R}^{w \times d}$ and $B_g \in \mathbb{R}^{d \times w}$. Its feed-forward block is

$$\begin{aligned} z_g &= A_g x, & \rho_g &= \sqrt{\|z_g\|_2^2 / w + \epsilon_{\text{act}}}, \\ h_g &= \rho_g \phi_g(z_g / \rho_g), & y &= \sum_{g=1}^G B_g h_g. \end{aligned} \quad (1)$$

The curve acts coordinatewise on RMS-normalized features and its output is multiplied by the same group scale (Zhang & Sennrich, 2019). The scale is computed per token and group, with stabilizer $\epsilon_{\text{act}} > 0$.

Each group learns a rational curve (Molina et al., 2020; Yang & Wang, 2025):

$$\phi_g(u) = \frac{\sum_{j=0}^5 a_{gj} u^j}{1 + \sum_{j=1}^4 |b_{gj}| |u|^j}. \quad (2)$$

The denominator is at least one, so the curve has no poles. All coefficients are shared across channels and token positions within a group and learned independently across groups. We initialize each curve with the same rational approximation to SiLU and train its coefficients jointly with the projection weights. Appendix B derives the Jacobian of the normalized activation and its dependence on input scale.

3.2 TILLER

TILLER (Tangent-Informed Loss Ledger for Equal-budget Reweighting) uses the learned activation response to construct matrix directions and allocate their strengths jointly. For an L -layer model, let $i = (\ell, g)$ index the $C = LG$ feed-forward groups. Each group receives incoming and outgoing directions (q_i^A, q_i^B) and a shared coefficient c_i . The directions determine how each group changes; the coefficients coordinate these changes across the model.

For either matrix in a pair, a calibrated polar direction p and a factorized adaptive direction a transform the same momentum (Jordan et al., 2024; Liu et al., 2025; Shazeer & Stern, 2018). Write $a^\perp = a - \langle a, p \rangle p / \|p\|_F^2$.

The incoming direction is guided by the activation Jacobian, and the outgoing direction by the activated features. For each role, loss-weighted participation $\pi \in [0, 1]$ measures the spread of this response. Congruence $\kappa \in [0, 1]$ compares its response kernel with that of the initial rational curve at the same current inputs. These summaries set the rotation:

$$\tilde{q} = \sqrt{1 - e} p + \sigma \sqrt{e} \frac{\|p\|_F}{\|a^\perp\|_F} a^\perp, \quad e = \pi(1 - \kappa^2), \quad (3)$$

where σ makes the tangent’s gradient contraction nonnegative. The rotation preserves the parent norm. An unchanged response kernel retains p ; broader responses permit larger rotations for a given departure from initialization. We retain p for a degenerate or nonfinite rotation, or when the parent p or candidate $\tilde{q}$ has nonpositive gradient contraction. Matrix-size scaling of the selected direction gives q . Appendix C defines the summaries, selection conditions, and norm-preserving geometry.

To coordinate these directions, periodic probes measure their loss responses at K example–token positions shared across layers. The probes also refresh π and κ ; between probes, directions are

recomputed from current moments using the latest response summaries. For block input $x_{k\ell}$, activated features h_{ki} , and activation Jacobian $J_{ki} = \partial h_{ki} / \partial z_{ki}$, the paired response is

$$S_{t,ki} = \langle \xi_{k\ell}, B_i J_{ki} q_i^A x_{k\ell} + q_i^B h_{ki} \rangle, \quad (4)$$

where $\xi_{k\ell}$ is the derivative of the summed minibatch loss at the block output, multiplied by the gradient-clipping factor. The combined response is $S_t c$: its squared norm includes cross terms between different groups and layers. Appendix D derives this differential and its relation to current-gradient descent.

A response matrix contributes the Gram metric $S_t^\top S_t / K$, whose off-diagonal entries measure interactions between pairs. The response a_t^{dec} to feed-forward weight decay contributes the cross term $S_t^\top a_t^{\text{dec}} / K$. Their discounted histories supply the current metric G_t and cross term r_t . Between probes, current gradient contractions supply repeated rows calibrated to the latest measured response scale. A Robust Frequent Directions sketch retains this history as a row factor and an isotropic remainder (Ghashami et al., 2016; Luo et al., 2019). The newest rows enter the metric before compression. Appendix D.1 gives the ledger construction and its approximation bounds.

Let g_i^A, g_i^B be the accumulated, averaged, clipped gradients, $s_{t,i} = \langle g_i^A, q_i^A \rangle + \langle g_i^B, q_i^B \rangle$ their descent score, and $\omega_i = \|q_i^A\|_F^2 + \|q_i^B\|_F^2$. The allocation balances current descent against the joint loss response:

$$F_t(c) = -\eta_t s_t^\top c + \frac{\eta_t^2}{2} (c^\top G_t c + 2r_t^\top c), \quad (5)$$

$$\sum_i \omega_i c_i^2 = \sum_i \omega_i,$$

where η_t is the learning rate. The linear term rewards current descent; the quadratic term penalizes the joint response of the directions and feed-forward weight decay. The constraint preserves the combined feed-forward direction norm of $c = 1$, before decay, while permitting signed coefficients. A coefficient proposal is used when it improves this model and satisfies the norm and layerwise contraction conditions; otherwise $c = 1$. Appendix E gives the projected solver and acceptance conditions.

With decay rate λ_{wd} , the resulting step applies

$$A_i \leftarrow (1 - \eta_t \lambda_{\text{wd}}) A_i - \eta_t c_i q_i^A, \quad B_i \leftarrow (1 - \eta_t \lambda_{\text{wd}}) B_i - \eta_t c_i q_i^B.$$

The coefficients jointly allocate these feed-forward pairs. An attention matrix D uses a response-guided direction q^D formed from its own moments and layerwise GRAIN summaries, with update $D \leftarrow (1 - \eta_t \lambda_{\text{wd}}) D - \eta_t q^D$. Rational coefficients and all remaining parameters use AdamW (Loshchilov & Hutter, 2019). Tables 1 and 2 collect the notation and the complete training loop.

3.2.1 FULL-HORIZON TILLER

For training horizon T , full-horizon TILLER applies this common update at every step. It continually adapts the matrix directions and their joint allocation as GRAIN’s nonlinearities evolve.

3.2.2 STAGED TRAINING

For a chosen transition $T_s < T$, staged training applies the same TILLER update through T_s and then continues the feed-forward and attention updates with an optimizer $\mathcal{U}$. Its state initializer $\mathcal{I}$ transfers stored first-moment histories compatible with the continuation’s recurrence and matrix grouping; additional statistics use $\mathcal{U}$ ’s initialization. The first continuation update acts on the learned matrices at $T_s + 1$. Model parameters and the learning-rate schedule continue across the transition. Rational coefficients and remaining parameters retain their AdamW state, and GRAIN remains trainable. Response collection and ledger updates end at T_s .

4 EXPERIMENTS

[TODO: Experiment section]

5 DISCUSSION

6 CONCLUSION

AI USE STATEMENT

Generative AI tools assisted with method development, experimental planning, implementation, mathematical derivations and proof checking, literature analysis, result interpretation, and manuscript drafting and revision.

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A NOTATION AND ALGORITHM

Table 1 summarizes the quantities used by TILLER. Table 2 combines the shared update with its full-horizon and staged training protocols. The subsequent sections develop the activation differential, direction geometry, response history, and coefficient allocation in this order.

Table 1: Principal notation and response expressions. The index $i = (\ell, g)$ identifies one of $C = LG$ feed-forward pairs, and k identifies one of K sampled example–token positions. Superscripts A and B distinguish incoming and outgoing matrix roles.

Quantity	Interpretation
π, κ	π measures loss-weighted response participation; κ compares the response kernel with that of the initial rational curve at the same current inputs.
$\tilde{q}$	Norm-preserving rotated candidate, before selection and matrix-size scaling.
q	Selected, matrix-size-scaled direction; feed-forward directions are subsequently multiplied by c_i .
c_i	Signed coefficient shared by the incoming and outgoing directions of pair i .
$B_i J_{ki} q_i^A x_{k\ell}$	Incoming-matrix contribution to the block-output differential: the preactivation change passes through the activation Jacobian and outgoing projection.
$q_i^B h_{ki}$	Outgoing-matrix contribution to the block-output differential, acting directly on the current activated features.
$\xi_{k\ell}$	Block-output derivative of the summed minibatch loss, multiplied by the gradient-clipping factor.
$S_t \in \mathbb{R}^{K \times C}$	Paired loss responses. On probes, each entry contracts the sum of the two output differentials with $\xi_{k\ell}$; between probes, rows are modeled from current-gradient contractions.
$S_t c \in \mathbb{R}^K$	Combined loss-response vector after weighting all feed-forward pairs by c , before adding the decay response.
$s_{t,i}$	Descent score of pair i at unit allocation, computed from the current accumulated, averaged, clipped minibatch gradients.
$a_t^{\text{dec}} \in \mathbb{R}^K$	Response to feed-forward weight decay, using the same measured or modeled construction as the paired responses.
$r_t \in \mathbb{R}^C$	Discounted cross term between the paired responses and the feed-forward decay response.
G_t	Metric used for the current allocation: discounted retained metric plus the Gram contribution of current response rows, before compression.
$\hat{G}_t$	Compressed response metric retained after step t for subsequent updates.
G_t^*	Uncompressed reference metric formed from measured and modeled responses, used to analyze approximation error.
ω_i	Paired squared direction norm $\ q_i^A\ _{\mathbb{F}}^2 + \ q_i^B\ _{\mathbb{F}}^2$; the combined feed-forward direction has squared norm $\sum_i \omega_i c_i^2$ before learning-rate scaling and decay.

Table 2: TILLER training. Set $T_s = T$ for full-horizon training, or choose $T_s < T$ and continuation optimizer $\mathcal{U}$ for staged training. Probe at the first step and periodically thereafter; reuse the latest response summaries between probes. Initialize model parameters, zero gradient moments, and $\hat{G}_0 = 0, r_0 = 0$.

- 1 For $t = 1, \dots, T$, compute the minibatch loss and clipped gradients g_t .
- 2 If $t > T_s$, initialize continuation state with $\mathcal{I}$ at $T_s + 1$, update the feed-forward and attention matrices with $\mathcal{U}$, and proceed to line 9.
- 3 Update matrix moments. On probes, evaluate features h , Jacobians J , loss-weighted participation π , and congruence κ with the initial activation at the current inputs. Form polar parents p , adaptive directions a , and tangents $a^\perp = a - \langle a, p \rangle p / \|p\|_F^2$.
- 4 Set $e = \pi(1 - \kappa^2)$ for feed-forward roles and $e = \sqrt{\pi^A \pi^B}(1 - \kappa^A \kappa^B)$ for attention, using layerwise summaries. Choose σ for a nonnegative tangent–gradient contraction and form $\tilde{q} = \sqrt{1 - e} p + \sigma \sqrt{e} \|p\|_F a^\perp / \|a^\perp\|_F$. Retain p unless the rotation is finite and nondegenerate and both $\langle g, p \rangle$ and $\langle g, \tilde{q} \rangle$ are positive. Apply matrix-size scaling to obtain q .
- 5 Set $s_i = \langle g_i^A, q_i^A \rangle + \langle g_i^B, q_i^B \rangle$ and $\omega_i = \|q_i^A\|_F^2 + \|q_i^B\|_F^2$. On probes, measure $S_{ki} = \langle \xi_{k\ell}, B_i J_{ki} q_i^A x_{k\ell} + q_i^B h_{ki} \rangle$; between probes, repeat the current-gradient row at the latest measured response scale. For a^{dec} , apply the same measured or current-gradient response model to directions $(\lambda_{\text{wd}} A_i, \lambda_{\text{wd}} B_i)$ and sum over pairs.
- 6 Use $\delta_t = 0$ at $t = 1$ and $\delta_t = \beta$ afterward. Form $G_t = \delta_t \hat{G}_{t-1} + (1 - \delta_t) S_t^\top S_t / K$ and $r_t = \delta_t r_{t-1} + (1 - \delta_t) S_t^\top a_t^{\text{dec}} / K$.
- 7 Propose c in a Krylov subspace to reduce $F_t(c) = -\eta_t s_t^\top c + \frac{\eta_t^2}{2} (c^\top G_t c + 2r_t^\top c)$ under $\sum_i \omega_i c_i^2 = \sum_i \omega_i$. Use $c = 1$ unless the proposal is finite, improves F_t , meets the norm tolerance, and both allocations have positive layer–role gradient and momentum contractions.
- 8 Apply $(A_i, B_i) \leftarrow (1 - \eta_t \lambda_{\text{wd}})(A_i, B_i) - \eta_t c_i (q_i^A, q_i^B)$ and $D \leftarrow (1 - \eta_t \lambda_{\text{wd}})D - \eta_t q^D$ for each attention matrix D . Retain the compressed metric $\hat{G}_t$ using Robust Frequent Directions.
- 9 Update rational coefficients and remaining parameters with AdamW.

B DIFFERENTIAL PROPERTIES OF GRAIN

For one feature group, write $\Phi(z) = \rho\phi(u)$, where $u = z/\rho$ and $\rho = (z^\top z/w + \epsilon_{\text{act}})^{1/2}$. The coefficients of ϕ are fixed when taking derivatives with respect to z . Differentiating the scale and normalized input gives

$$\nabla_z \rho = u/w, \quad \frac{\partial u}{\partial z} = \rho^{-1}(I - uu^\top/w). \quad (6)$$

Applying the product rule to $\rho\phi(u)$ then yields

$$J = \text{Diag}(\phi'(u)) + \frac{[\phi(u) - u \odot \phi'(u)]u^\top}{w}. \quad (7)$$

The first term is the coordinatewise rational slope. The second is the response through the shared normalization scale. At the absolute-value kinks, we use the selected derivative $\text{sign}(0) = 0$.

This expression distinguishes changes in feature shape from changes in amplitude. A perturbation orthogonal to u leaves ρ unchanged to first order, so only the diagonal term acts. For a radial perturbation, set $r_\phi = \phi(u) - u \odot \phi'(u)$ and use $\|u\|_2^2/w = 1 - \epsilon_{\text{act}}/\rho^2$ to obtain

$$Jz = \Phi(z) - \frac{\epsilon_{\text{act}}}{\rho} r_\phi. \quad (8)$$

In particular, when $\epsilon_{\text{act}} = 0$ and $z \neq 0$, $\Phi(az) = a\Phi(z)$ for every $a > 0$. The rational curve changes the response shape while positive input rescaling is carried to the output. Equation 8 gives the correction for a positive stabilizer. For a linear curve $\phi(u) = au$, $r_\phi = 0$ and $\Phi(z) = az$ for every stabilizer value.

The two derivative actions used in TILLER follow from the same diagonal-plus-rank-one form:

$$Jv = \phi'(u) \odot v + r_\phi \frac{u^\top v}{w}, \quad J^\top \zeta = \phi'(u) \odot \zeta + u \frac{r_\phi^\top \zeta}{w}. \quad (9)$$

Thus both the forward response to a perturbation and the backward loss sensitivity retain the effect of normalization. Each action uses one inner product in addition to coordinatewise products.

C GEOMETRY OF RESPONSE-GUIDED DIRECTIONS

TILLER constructs a polar parent and an adaptive tangent from the same momentum, then uses the activation response to set the rotation between them. For a clipped matrix gradient g_t , the stored first moment and the Nesterov momentum are

$$\bar{m}_t = \beta_m \bar{m}_{t-1} + (1 - \beta_m) g_t, \quad m_t = (1 - \beta_m) g_t + \beta_m \bar{m}_t.$$

The history $\bar{m}_t$ is the state carried to a compatible continuation optimizer in staged training.

The adaptive direction uses row and column sums R_t, C_t of the exponentially averaged squared gradients. Summation commutes with exponential averaging, so these marginals have the recurrences $R_t = \beta_2 R_{t-1} + (1 - \beta_2)(g_t \odot g_t) \mathbf{1}$ and $C_t = \beta_2 C_{t-1} + (1 - \beta_2)(g_t \odot g_t)^\top \mathbf{1}$. With zero-initialized moments, the adaptive transformation is

$$V_t = \frac{R_t C_t^\top}{(\mathbf{1}^\top R_t)(1 - \beta_2^t)}, \quad a_t = \frac{m_t}{\sqrt{V_t} + \epsilon_{\text{opt}}}. \quad (10)$$

All operations in the second expression are entrywise. For positive total second moment, the normalized outer product is the rank-one approximation determined by its row and column marginals (Shazeer & Stern, 2018).

The group calibration follows from the norm of an exact polar factor. The parent is $p = \chi \mathcal{P}_5(m)$, where $\mathcal{P}_5$ is the five-iteration Newton–Schulz polar approximation (Jordan et al., 2024; Liu et al., 2025). For G groups of width w and total intermediate width $H_r = Gw$, $\chi = \sqrt{\min(H_r, d)/(G \min(w, d))}$. An exact full-rank polar factor of a group matrix has squared Frobenius norm $\min(w, d)$; hence this calibration makes the sum over groups equal the whole-projection value $\min(H_r, d)$. The selected feed-forward direction also receives the usual whole-matrix multiplier $\mu = 0.2 \sqrt{\max(H_r, d)}$.

The rotation is specified by the learned response geometry. For nonzero J and h , define

$$\text{pr}_A(J) = \frac{\text{tr}(J J^\top)^2}{w \|J J^\top\|_F^2}, \quad \text{pr}_B(h) = \frac{\|h\|_2^4}{w \|h\|_4^4}. \quad (11)$$

Both are normalized participation ratios. The first applies to squared singular values of the Jacobian and the second to squared feature coordinates. Equal energy in m nonzero components gives m/w ; the ratios lie in $[1/w, 1]$ by Cauchy–Schwarz. Their weighted averages, denoted π_i^A and π_i^B , use $\alpha_{ki}^A = \|J_{ki}^\top B_i^\top \xi_{k\ell}\|_2^2/w$ and $\alpha_{ki}^B = \|\xi_{k\ell}\|_2^2/d$:

$$\pi_i^r = \frac{\sum_k \alpha_{ki}^r \text{pr}_r(\mathcal{R}_{ki}^r)}{\sum_k \alpha_{ki}^r}, \quad \mathcal{R}_{ki}^A = J_{ki}, \quad \mathcal{R}_{ki}^B = h_{ki}. \quad (12)$$

Zero responses contribute zero participation. The vector $J_{ki}^\top B_i^\top \xi_{k\ell}$ is the loss sensitivity at the group preactivation; $\xi_{k\ell}$ is the sensitivity at the block output. Their squared norms weight the incoming and outgoing paths. When either role has zero total weight, both group participations are set to one.

To compare the learned response with its initialization, define $\Gamma_{ki}^A = J_{ki} J_{ki}^\top$ and $\Gamma_{ki}^B = h_{ki} h_{ki}^\top$. The first Gram matrix records inner products between rows of the activation Jacobian; the second records products of activated feature coordinates. They describe the response patterns compared with initialization. A superscript 0 denotes the same evaluation with initial rational coefficients, at the current inputs and scales. For either role,

$$\kappa_i^r = \frac{\sum_k \alpha_{ki}^r \langle \Gamma_{ki}^r, \Gamma_{ki}^{r,0} \rangle}{\sqrt{\sum_k \alpha_{ki}^r \|\Gamma_{ki}^r\|_F^2} \sqrt{\sum_k \alpha_{ki}^r \|\Gamma_{ki}^{r,0}\|_F^2}}. \quad (13)$$

This is a cosine in the space of weighted kernel collections. Positive semidefiniteness and Cauchy–Schwarz give $0 \leq \kappa_i^r \leq 1$. If either congruence denominator vanishes, both group congruences are set to one, giving zero feed-forward rotation. Using the same current inputs in both collections isolates the change in the learned activation. If $\psi = \arccos \kappa$, the fraction $e = \pi(1 - \kappa^2) = \pi \sin^2 \psi$ combines the kernel angle with the breadth of its response.

For a nonzero parent and tangent, the corresponding matrix rotation is

$$a^\perp = a - \frac{\langle a, p \rangle}{\|p\|_F^2} p, \quad \tilde{q} = \sqrt{1-e} p + \sigma \sqrt{e} \frac{\|p\|_F}{\|a^\perp\|_F} a^\perp, \quad (14)$$

where σ makes $\sigma \langle g, a^\perp \rangle \geq 0$. Orthogonality gives

$$\|\tilde{q}\|_F^2 = \|p\|_F^2, \quad \langle g, \tilde{q} \rangle = \sqrt{1-e} \langle g, p \rangle + \sqrt{e} \frac{\|p\|_F}{\|a^\perp\|_F} |\langle g, a^\perp \rangle|. \quad (15)$$

The angle ϑ between p and $\tilde{q}$ therefore satisfies $\sin^2 \vartheta = \pi \sin^2 \psi$. Broad responses admit larger rotations for the same kernel departure; an unchanged kernel gives zero rotation. The common multiplier μ preserves the angle and multiplies both squared norms by μ^2 . The rotated candidate is selected when the parent and tangent are nonzero, the computed quantities are finite, and both $\langle g, p \rangle$ and $\langle g, \tilde{q} \rangle$ are positive. Otherwise the parent is retained.

For attention, the polar parent is assembled from head groups. Apply $\mathcal{P}_5$ separately to each group of rows in the query, key, and value projections and to each corresponding group of columns in the output projection. The stacked query–key–value parent is scaled by $1/\sqrt{3}$; the output parent is unscaled. The adaptive direction and tangent are then formed for each assembled matrix using its own moments. For a matrix of size $n \times m$, the final direction multiplier is $0.2\sqrt{\max(n, m)}$.

The attention rotation uses the layer’s two response roles jointly. Let $\bar{\pi}_\ell^A, \bar{\pi}_\ell^B$ be group-averaged participation, and let $\kappa_\ell^A, \kappa_\ell^B$ use kernel inner products and squared norms aggregated across groups.

Its rotation fraction is $e_\ell^{\text{att}} = \sqrt{\bar{\pi}_\ell^A \bar{\pi}_\ell^B} (1 - \kappa_\ell^A \kappa_\ell^B)$. All four role statistics lie in $[0, 1]$, so this layerwise fraction also lies in $[0, 1]$. It is set to zero when either aggregated congruence denominator vanishes.

These summaries are refreshed on probe steps. Between probes, the moments and directions continue to change at every update while the latest response summaries determine the rotation fractions.

D THE JOINT LOSS-RESPONSE MODEL

Consider the block $B_i \Phi_i(A_i x_{k\ell})$ at a fixed position. The derivative along its paired matrix direction is

$$\left. \frac{d}{d\alpha} (B_i + \alpha q_i^B) \Phi_i((A_i + \alpha q_i^A) x_{k\ell}) \right|_{\alpha=0} = B_i J_{ki} q_i^A x_{k\ell} + q_i^B h_{ki}. \quad (16)$$

Contracting with the downstream loss derivative gives $S_{t,ki}$ in equation 4. The derivative at a hidden position collects its effect on the entire downstream loss. With microbatch mean loss over N positions and M accumulation steps, the captured derivative ξ has scale $1/(NM)$; thus $\xi = NM \chi_{\text{clip}} \bar{\xi}$, where χ_{clip} is the realized gradient-clipping factor, held fixed in this calculation. Averaging over all positions and processes recovers the clipped directional derivative $s_i = \langle g_i^A, q_i^A \rangle + \langle g_i^B, q_i^B \rangle$.

The full gradient determines average descent. Shared-position measurements additionally determine how directions act together. For columns S_i and S_j , their contribution to the squared joint response is $2c_i c_j S_i^\top S_j / K$. For the scaled directions, the sign of $c_i c_j S_i^\top S_j$ determines reinforcement or cancellation. These cross terms couple groups even when they occupy disjoint parameter blocks. Weight decay has the same differential: replacing the directions by $(\lambda_{\text{wd}} A_i, \lambda_{\text{wd}} B_i)$ and summing over i gives a response a^{dec} . Expanding the combined response,

$$\frac{\|Sc + a^{\text{dec}}\|_2^2}{K} = c^\top \frac{S^\top S}{K} c + 2c^\top \frac{S^\top a^{\text{dec}}}{K} + \frac{\|a^{\text{dec}}\|_2^2}{K}. \quad (17)$$

The final term is independent of the allocation. The first two are the response metric and decay cross term in F_t . Their discounted history supplies the quadratic penalty, while the current s_t supplies the linear descent term.

D.1 APPROXIMATION OF THE RESPONSE HISTORY

Let S_t denote the measured rows on a probe step. Between probes, the row model is $S_t = \mathbf{1}_K (\gamma_t s_t)^\top$, where $\gamma_t = \nu / \|s_t\|_2$ for nonzero s_t and $\nu = \|S_{\text{ref}}\|_F / \sqrt{K}$ is the latest measured scale. This model

has $S_t^\top S_t/K = \nu^2 s_t s_t^\top / \|s_t\|_2^2$: its orientation follows the current gradient contractions and its trace equals the measured ν^2 . For $s_t = 0$, set $\gamma_t = 0$, giving zero rows. The corresponding decay rows are $a_t^{\text{dec}} = \mathbf{1}_K \gamma_t d_t^{\text{dec}}$, with $d_t^{\text{dec}} = \lambda_{\text{wd}} \sum_i (\langle g_i^A, A_i \rangle + \langle g_i^B, B_i \rangle)$.

For the resulting sequence of measured and modeled responses, with the second-moment decay $\beta = \beta_2$, define

$$G_t^* = \beta G_{t-1}^* + (1 - \beta) S_t^\top S_t / K, \quad G_1^* = S_1^\top S_1 / K. \quad (18)$$

Each column retains its layer-group identity; each historical row records the direction used at its own training step. The ledger approximates this metric by $\hat{G}_t = B_t^\top B_t / K + \tau_t I$. If $\mathcal{A}_t = [\sqrt{\beta} B_{t-1}; \sqrt{1 - \beta} S_t]$, the metric for the current allocation is

$$G_t = \mathcal{A}_t^\top \mathcal{A}_t / K + \beta \tau_{t-1} I, \quad r_t = \beta r_{t-1} + (1 - \beta) S_t^\top a_t^{\text{dec}} / K. \quad (19)$$

The first observation is unattenuated: $A_1 = B_1 = S_1$, $\tau_1 = 0$, and $r_1 = S_1^\top a_1^{\text{dec}} / K$. Fresh rows enter G_t before compression.

The Robust Frequent Directions approximation (Ghashami et al., 2016; Luo et al., 2019) has a direct spectral description. Let σ_j and v_j be the singular values and right singular vectors of $\mathcal{A}_t$. For a retained rank r_{sk} , set $\Delta_t = \sigma_{r_{\text{sk}}+1}^2$, or zero if that singular value is absent. The retained factor has rows $\sqrt{\sigma_j^2 - \Delta_t} v_j^\top$ for the available $j \leq r_{\text{sk}}$, and $\tau_t = \beta \tau_{t-1} + \Delta_t / (2K)$. The isotropic increment centers the error from spectral shrinkage.

Proposition 1 (Error in the response metric). *For finite response rows, $K > 0$, $r_{\text{sk}} \geq K$, $0 \leq \beta \leq 1$, and $\tau_{t-1} \geq 0$, $\hat{G}_t$ is positive semidefinite and*

$$\|\hat{G}_t - G_t\|_2 \leq \frac{\Delta_t}{2K}, \quad |c^\top (\hat{G}_t - G_t) c| \leq \frac{\Delta_t}{2K} \|c\|_2^2. \quad (20)$$

Proof. The factor Gram matrix and isotropic term are positive semidefinite. In the right-singular-vector basis, a retained direction loses Δ_t/K by shrinkage and gains $\Delta_t/(2K)$, giving error $-\Delta_t/(2K)$. A discarded direction with squared singular value $s \in [0, \Delta_t]$ has error $(\Delta_t/2 - s)/K$. The same expression with $s = 0$ covers null directions. Every error eigenvalue therefore lies in $[-\Delta_t/(2K), \Delta_t/(2K)]$, proving both bounds. $\square$

Without the isotropic increment, the error interval is $[-\Delta_t/K, 0]$. Centering halves this one-compression spectral bound. Let $E_t = \hat{G}_t - G_t^*$. Since $E_t = \beta E_{t-1} + (\hat{G}_t - G_t)$ and $E_1 = 0$,

$$\|E_t\|_2 \leq \sum_{j=2}^t \beta^{t-j} \frac{\Delta_j}{2K}, \quad G_t - G_t^* = \beta E_{t-1}. \quad (21)$$

This connects the sketch approximation to the coefficient model. If $\|G_t - G_t^*\|_2 \leq \varepsilon_t$, replacing G_t by G_t^* changes $F_t(c)$ by at most $\eta_t^2 \varepsilon_t \|c\|_2^2 / 2$. Under the fixed-norm constraint and positive ω_i , this is at most $\eta_t^2 \varepsilon_t \mathcal{E}_0 / (2 \min_i \omega_i)$, where $\mathcal{E}_0 = \sum_i \omega_i$.

E THE FIXED-NORM ALLOCATION PROBLEM

Consider prescribed paired directions (q_i^A, q_i^B) . Their parameter blocks are disjoint, so the feed-forward update before weight decay has norm

$$\|\Delta\theta(c)\|_2^2 = \eta_t^2 \sum_i c_i^2 (\|q_i^A\|_F^2 + \|q_i^B\|_F^2) = \eta_t^2 c^\top W c, \quad W = \text{Diag}(\omega). \quad (22)$$

The constraint $c^\top W c = \mathcal{E}_0$, where $\mathcal{E}_0 = \mathbf{1}^\top W \mathbf{1}$, therefore fixes the direction norm at its unit-allocation value. A shared coefficient preserves the relative scaling of each group’s incoming and outgoing directions, including when that coefficient is negative.

For positive ω_i and η_t , substitute $y = W^{1/2} c$ in the objective. This gives

$$\begin{aligned} \frac{F_t(c)}{\eta_t^2} &= Q_t(y) = \frac{1}{2} y^\top M_t y - b_t^\top y, \quad \|y\|_2^2 = \mathcal{E}_0, \\ M_t &= W^{-1/2} G_t W^{-1/2}, \quad b_t = W^{-1/2} (s_t / \eta_t - r_t). \end{aligned} \quad (23)$$

Differentiating the Lagrangian $\mathcal{Q}_t(y) + \lambda_c(\|y\|_2^2 - \mathcal{E}_0)/2$ yields $(M_t + \lambda_c I)y = b_t$. The positive semidefiniteness of G_t makes $M_t + \lambda_c I$ positive definite for $\lambda_c > 0$.

TILLER represents this relation in a Lanczos-generated Krylov subspace (Lanczos, 1950). For $b_t \neq 0$, the starting vector is $b_t/\|b_t\|_2$, with orthogonal completion on numerical breakdown. Let Q_k be the resulting orthonormal basis and let T_k be the Lanczos tridiagonal model. Clipping negative numerical eigenvalues gives $T_k^+ = U \text{Diag}(\theta)U^\top$. For $\gamma = U^\top Q_k^\top b_t$, projected stationarity gives

$$y = Q_k U \text{Diag}((\theta + \lambda_c)^{-1})\gamma, \quad \psi(\lambda_c) = \sum_j \frac{\gamma_j^2}{(\theta_j + \lambda_c)^2} = \mathcal{E}_0. \quad (24)$$

The scalar equation enforces the norm of the projected point. If its positive-multiplier branch does not yield a norm-feasible proposal, TILLER retains $c = 1$. The positive branch has the following characterization.

Proposition 2 (Positive-multiplier allocation). *Let $T_k^+ \succeq 0$, $Q_k^\top Q_k = I$, and $\gamma \neq 0$. Equation 24 has at most one root with $\lambda_c > 0$. If such a root exists, its projected coordinates are the unique minimizer of the clipped projected quadratic on the sphere of squared radius $\mathcal{E}_0$.*

Proof. For positive λ_c , $\psi'(\lambda_c) = -2 \sum_j \gamma_j^2 / (\theta_j + \lambda_c)^3 < 0$, which proves uniqueness of the root. Put $b_k = Q_k^\top b_t$ and $z_* = (T_k^+ + \lambda_c I)^{-1} b_k$. For any z on the same sphere, stationarity and equality of norms give

$$\left(\frac{1}{2}z^\top T_k^+ z - b_k^\top z\right) - \left(\frac{1}{2}z_*^\top T_k^+ z_* - b_k^\top z_*\right) = \frac{1}{2}(z - z_*)^\top (T_k^+ + \lambda_c I)(z - z_*). \quad (25)$$

The right side is positive unless $z = z_*$. $\square$

The mapped coefficients $c = W^{-1/2}y$ supply a proposal for F_t . Its improvement is evaluated with the current metric G_t ; acceptance also requires the prescribed norm tolerance.

Norm preservation permits sign changes within a layer. Their aggregate effect on a layer and matrix role is measured by

$$\sum_g c_{\ell g} \langle g_{\ell g}^r, q_{\ell g}^r \rangle, \quad \sum_g c_{\ell g} \langle m_{\ell g}^r, q_{\ell g}^r \rangle, \quad r \in \{A, B\}. \quad (26)$$

Requiring these contractions to be positive for both the proposal and the unit allocation constrains the whole incoming and outgoing updates, while allowing individual group coefficients to be negative.